Rajapu Praveen Naidu

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	7.79
Tirumala Educational Institutes	Class XII (BIEAP)	2024	95.30%
Geetanjali School	Class X (BSEAP(SSC))	2022	93.33%

SKILLS

Languages: C++, C, Python, SQL, JavaScript, HTML, CSS

Frameworks & Libraries: FastAPI, React, SQLAlchemy, Jinja2

Databases: PostgreSQL, SQLite

Tools & Platforms: Git, GitHub, Linux, Vercel, Render, Neon PostgreSQL

Backend & Systems: REST APIs, JWT Authentication, Session Authentication, Socket Programming, TCP/IP, Multithreading, Client-Server Architecture

COMPETITIVE PROGRAMMING

- Solved 700+ algorithmic problems across LeetCode (350+) and Codeforces (300+).
- Developed strong algorithmic thinking through extensive practice in Dynamic Programming, Graph Algorithms, Trees, Greedy Algorithms, and Binary Search.

RELEVANT COURSES

Computer Science: Data Structures and Applications,Algorithms, DBMS, OS, Theory of computation,Discrete Mathematics

Mathematics: Calculus-1&2, Elementary Linear Algebra, Differential equation, Transform Techniques, Statistics,

Complex Variables, Probability and Random Variables

PROJECTS

FileVault | React | FastAPI | PostgreSQL

- Developed a secure cloud file management platform using React, FastAPI, PostgreSQL, and SQLAlchemy, implementing JWT-based authentication, RESTful APIs, and secure file storage.
- Implemented end-to-end file lifecycle management supporting upload, download, rename, deletion, search, favorites, and storage analytics through RESTful backend services.
- Deployed the application on Vercel, Render, and Neon PostgreSQL while configuring production authentication, CORS policies, and environment-specific deployment workflows.

Multi-threaded TCP Key-Value Store | C++ | Systems Programming

- Developed a Redis-inspired key-value database from scratch in C++ with a custom TCP client-server architecture supporting concurrent client requests.
- Implemented an in-memory database engine with commands including SET, GET, DEL, EXISTS, TTL-based expiration, and file persistence.
- Built a concurrent request-handling system using multithreading/thread pool concepts, socket programming, and backend request processing workflows.

Unix-like Shell | C++ | Linux System Programming

- Built a Unix-like shell in C++ using Linux system calls, supporting command execution, built-in commands, I/O redirection, pipelines, and background process execution.
- Implemented core OS concepts including process creation (fork), program execution (execvp), process synchronization (wait/waitpid), pipes, file descriptors, and signal handling.
- Designed command execution flow with support for multi-stage pipelines, asynchronous processes, and zombie process cleanup using waitpid() and Linux system calls.

Expense Tracker | FastAPI | SQLAlchemy | SQLite

- Built a full-stack expense tracker with session-based authentication and bcrypt password hashing, using FastAPI, SQLAlchemy, and Jinja2 templates
- Implemented full CRUD for expenses (add, edit, delete) with combined filtering by category, month, year, and title search on the same dashboard
- Computed spending analytics -- total, average, and per-category breakdowns across 4 categories -- rendered server-side through Jinja2

EXTRACURRICULAR

- Member of the NSO Cricket team